10.28 Feynman Rules for Dirac Fields

LecNote19 works one process through the LSZ formula, $e^-\varphi \to e^-\varphi$ in Yukawa theory (pp. 4–6), starts $e^+\varphi \to e^+\varphi$ (p. 7) and announces "We can see the 'Rules' here". LecNote20 adds the arrow on fermion lines and more trees (pp. 1–3); the rules themselves are listed only briefly, in the note on LecNote19 p. 7. Page numbers without a LecNote refer to LecNote19. Students ask:

- "Which spinor goes where for an incoming positron, and on which side of the matrix?"
- "Can I copy p. 6 for $e^+\varphi$ with $u \to v$? What is the 'flipped sign in p' on p. 7?"
- "Fermions anticommute. Which minus signs must I track, and which can I drop?"

In short: LSZ attaches u to an incoming electron, $\bar{u}$ to an outgoing one, $\bar{v}$ to an incoming positron and v to an outgoing one. A fermion line carries an arrow from its $\bar{\Psi}$ end to its Ψ end and is written from the head of the arrow backward. A positron's momentum runs against the arrow, so copying p. 6 gets the numerators wrong. Relative signs between diagrams, and the -1 of a closed loop, are physical; the overall sign of iT is a convention, and the rules leave it free.

Conventions are those of §10.26: metric $(-,+,+,+)$, $\{\gamma^\mu,\gamma^\nu\} = -2g^{\mu\nu}$, $\bar{\Psi} = \Psi^\dagger\beta$, $\mathcal{L} = i\bar{\Psi}\partial\!\!\!/\Psi - m\bar{\Psi}\Psi - \frac{1}{2}\partial\varphi\partial\varphi - \frac{1}{2}M^2\varphi^2 + g\varphi\bar{\Psi}\Psi$ (M is the m_φ of LecNote19–20), $(p\!\!\!/ + m)u_s(\mathbf{p}) = 0$, $(-p\!\!\!/ + m)v_s(\mathbf{p}) = 0$, $\sum_s u_s\bar{u}_s = -p\!\!\!/ + m$, $\sum_s v_s\bar{v}_s = -p\!\!\!/ - m$. The propagators are $\langle 0|T\Psi(x)\bar{\Psi}(y)|0\rangle = \frac{1}{i}S(x-y)$ and $\langle 0|T\varphi(x)\varphi(y)|0\rangle = \frac{1}{i}\Delta(x-y)$; with $S(x) = \int \frac{d^4p}{(2\pi)^4}e^{ipx}\tilde{S}(p)$ and likewise for Δ, $\tilde{S}(p) = (-p\!\!\!/ + m)/(p^2 + m^2 - i\epsilon)$ and $\tilde{\Delta}(k) = 1/(k^2 + M^2 - i\epsilon)$ (LecNote19 pp. 3–4). As $(p\!\!\!/ + m)(-p\!\!\!/ + m) = p^2 + m^2$, $\tilde{S}(p)$ inverts $p\!\!\!/ + m$ from either side. As in (10.11.4), $\langle f|i\rangle = (2\pi)^4\delta^4(\sum p_{\text{in}} - \sum p_{\text{out}})\, iT$.

LSZ for spin one-half

The mode operators $b_s^\dagger(\mathbf{p}) = \int d^3x\, e^{ipx}\bar{\Psi}\gamma^0 u_s$ and $d_s^\dagger(\mathbf{p}) = \int d^3x\, e^{ipx}\bar{v}_s\gamma^0\Psi$ of LecNote18 p. 13, built from the interacting field, depend on t; ∂_0 of each is a space integral of the Dirac operator on the field (LecNote19 pp. 1–2). Time ordering sends the out creation and in annihilation operators to the

vacua (§10.6), so in $\langle f|i\rangle = \langle 0|T\cdots|0\rangle$

$$\begin{aligned} b_s^\dagger(\mathbf{p})_{\text{in}} &\to \quad i\int d^4x\ \bar\Psi(x)\,(+i\overleftarrow{\partial\!\!\!/}+m)\,u_s(\mathbf{p})\,e^{ipx},\\ b_s(\mathbf{p})_{\text{out}} &\to \quad i\int d^4x\ e^{-ipx}\,\bar u_s(\mathbf{p})\,(-i\partial\!\!\!/+m)\,\Psi(x),\\ d_s^\dagger(\mathbf{p})_{\text{in}} &\to -i\int d^4x\ e^{ipx}\,\bar v_s(\mathbf{p})\,(-i\partial\!\!\!/+m)\,\Psi(x),\\ d_s(\mathbf{p})_{\text{out}} &\to -i\int d^4x\ \bar\Psi(x)\,(+i\overleftarrow{\partial\!\!\!/}+m)\,v_s(\mathbf{p})\,e^{-ipx}, \end{aligned} \tag{10.28.1}$$

and $a_{\text{in}}^\dagger, a_{\text{out}} \to i\int d^4x\, e^{\pm ikx}(-\partial^2+M^2)\varphi$. (This fixes the signs in LecNote19 p. 1's scalar line and p. 2's $d^\dagger$ line.) So an incoming e^- or an outgoing e^+ sits at a $\bar\Psi$ point, an outgoing e^- or an incoming e^+ at a Ψ point: Ψ annihilates electrons and creates positrons.

Each external point is joined to a vertex w by a free propagator, which the Dirac operator inverts, $(-i\partial\!\!\!/_x+m)S(x-w)=\delta^4(x-w)=S(w-x)(+i\overleftarrow{\partial\!\!\!/}_x+m)$ (LecNote19 p. 3). The external point is set to w, and the LSZ factor times the $\frac{1}{i}$ of the propagator leaves

$$u\,e^{ipw}\ (e^-\text{ in}),\quad \bar u\,e^{-ipw}\ (e^-\text{ out}),\quad \bar v\,e^{ipw}\ (e^+\text{ in}),\quad v\,e^{-ipw}\ (e^+\text{ out}), \tag{10.28.2}$$

and $e^{\pm ikw}$ for a scalar: a spinor, and the momentum fed into (or taken out of) the vertex, like the factor 1 of §10.11. For a positron the prefactors give $(-i)\frac{1}{i}=-1$. Every diagram of a process has the same positrons, so this is one factor $(-1)^{N_{e^+}}$ in iT (N_{e^+} external positrons), the same as giving each positron state the opposite phase; it drops out of $|T|^2$, and the rules, like LecNote19 p. 7, leave it out. Loops on external legs are not drawn, as for scalars: LecNote19 p. 3 normalizes Ψ so that the pole of the exact propagator gives exactly these factors.

Vertices, lines and the arrow

The vertex is $ig\int d^4w\,\varphi\bar\Psi\Psi$; the $1/V!$ cancels against the namings of the vertices (LecNote18 p. 1, §10.9). One fermion line enters each vertex and one leaves ("one away, one toward"), so fermion lines run from external point to external point or close into loops. The generating functional $\exp[i\int\bar\eta S\eta]$ pairs a $\Psi(x)$ with a $\bar\Psi(y)$ through $S(x-y)$ (LecNote17 p. 15, LecNote20 p. 1): the arrow points from y to x, from the $\bar\Psi$ end to the Ψ end. It is the flow of the charge $Q=N_b-N_d$ (LecNote18 p. 9); electrons move along it, positrons against it.

An internal line is $\frac{1}{i}S(w_2-w_1)=\int\frac{d^4q}{(2\pi)^4}e^{iq(w_2-w_1)}\frac{1}{i}\tilde S(q)$, with q flowing from w_1 to w_2, along the arrow. Unlike $\tilde\Delta$, $\tilde S$ is not even: a line whose physical momentum p runs against the arrow carries $\tilde S(-p)=(\not p+m)/(p^2+m^2-i\epsilon)$,

LecNote19 p. 7's "the sign in p is flipped". The matrices multiply in the order of $S(x-w_2)S(w_2-w_1)S(w_1-y)$: start at the head of the arrow and walk back. The spinor at the head stands on the left, $\bar{u}'\cdots u$ for an electron line and $\bar{v}_{\text{in}}\cdots v_{\text{out}}$ for a positron line.

Signs

Each complete set of contractions in a correlator of Ψ's and $\bar{\Psi}$'s comes with the sign of the permutation that brings the fields into adjacent pairs $\Psi\bar{\Psi}$, each pair being S/i (§10.27). The bilinears $\bar{\Psi}\Psi$ commute with everything, so walking along an open line gives no sign, and graphs that differ only in where the φ's attach add with $+$; a closed fermion loop gives -1 and a trace. Graphs that pair the external fermions differently can differ in sign, and this is physical: read each chain from the head of its arrow to its tail and list the external labels in the order their spinors appear; two terms whose lists differ by an odd permutation differ by -1. For identical fermions this is Fermi statistics.

The overall sign of iT is a convention: reordering the operators in a state ($b_2^\dagger b_1^\dagger|0\rangle$ for $b_1^\dagger b_2^\dagger|0\rangle$), like the phase of the positron states, flips every diagram at once. If you keep every sign, a term's overall sign is $(-1)^{N_{e^+}}$ times the parity of the permutation from the order of the external fermion fields in $\langle f|i\rangle$ to the order of its spinors.

The rules

For iT, draw all diagrams with labelled external lines; then write

- ig for each vertex (a φ line, one fermion arrow in, one out), and 1 for each external φ;
- $u_s(\mathbf{p})$ for an incoming e^-, $\bar{u}_{s'}(\mathbf{p}')$ for an outgoing e^-, $\bar{v}_s(\mathbf{p})$ for an incoming e^+, $v_{s'}(\mathbf{p}')$ for an outgoing e^+ (whose momentum runs against the arrow);
- $\frac{1}{i}\tilde{S}(q)$ for an internal fermion line, q along the arrow, and $\frac{1}{i}\tilde{\Delta}(k)$ for an internal scalar; conserve momentum at each vertex, integrate $\int d^4l/(2\pi)^4$ over each loop momentum, and include symmetry factors as for scalars (arrows cannot be reversed, so a fermion loop on a φ line has $S=1$);
- each open line as a matrix chain from the head of its arrow to its tail, and each closed fermion loop as -1 times a trace;
- -1 between terms whose spinor orders differ by an odd permutation; the overall sign is free.

Worked examples

$e^-\varphi\to e^-\varphi$ (LecNote19 pp. 4–6). In $\langle 0|Ta(\mathbf{k}')b_{s'}(\mathbf{p}')b_s^\dagger(\mathbf{p})a^\dagger(\mathbf{k})|0\rangle$, whose fields stand as in LecNote18 p. 1's $\langle 0|T\Psi\bar{\Psi}\varphi\varphi|0\rangle_c$, LSZ's i^4 turns its $(\frac{1}{i})^5(ig)^2$ into

$\frac{1}{i}(ig)^2$. The φ's attach in either order; a t-channel graph would need a φ^3 vertex. With $s=-(p+k)^2$, $u=-(p-k')^2$, $t=-(p-p')^2$, $s+t+u=2m^2+2M^2$,

$$\begin{aligned} iT &= \tfrac{1}{i}(ig)^2\,\bar u_{s'}(\mathbf{p}')\big[\tilde S(p+k)+\tilde S(p-k')\big]u_s(\mathbf{p}),\\ T &= g^2\,\bar u_{s'}(\mathbf{p}')\Big[\frac{-\not p-\not k+m}{m^2-s}+\frac{-\not p+\not k'+m}{m^2-u}\Big]u_s(\mathbf{p}), \end{aligned} \tag{10.28.3}$$

as on LecNote20 pp. 2 and 4 (on LecNote19 p. 6 the second denominator should be $(p-k')^2$). For $M<2m$ (φ stable) no denominator in this section vanishes in the physical region, and the $-i\epsilon$ is dropped. The spin-summed $|T|^2$ is in §10.29.

$e^+\varphi\to e^+\varphi$ (LecNote19 p. 7). In $\langle 0|Ta(\mathbf{k}')d_{s'}(\mathbf{p}')d_s^\dagger(\mathbf{p})a^\dagger(\mathbf{k})|0\rangle$ the positron enters at a Ψ point x and leaves at a $\bar\Psi$ point y. The arrow runs from y to x, so the momentum along it is $-(p+k)$ or $-(p-k')$, and the chain starts with $\bar v$ at x:

$$iT=\tfrac{1}{i}(ig)^2\,\bar v_s(\mathbf{p})\Big[\frac{\not p+\not k+m}{m^2-s}+\frac{\not p-\not k'+m}{m^2-u}\Big]v_{s'}(\mathbf{p}'), \tag{10.28.4}$$

as LecNote19 p. 7's note writes it. (Keeping every sign gives an extra -1: $\bar\Psi(y)\Psi(x)=-\Psi(x)\bar\Psi(y)$ in T, and the two positron legs give $(-1)^2$.) Copying (10.28.3) with $u\to v$ fails twice: the spinors trade ends and the numerators change sign. (LecNote20 p. 2 keeps the electron numerators, which give a different spin-summed $|T|^2$.) A check: with LecNote18's phases for v ($v_s=C\bar u_s^{\,T}$, §10.26), $\bar v_s(\mathbf{p})(\not q+m)v_{s'}(\mathbf{p}')=-\bar u_{s'}(\mathbf{p}')(-\not q+m)u_s(\mathbf{p})$ for any q, so (10.28.4) is $-$(10.28.3) spin by spin (+ with every sign kept): the same $|T|^2$ for each pair of spins, as charge conjugation, a symmetry of $g\varphi\bar\Psi\Psi$, requires.

$e^-e^-\to e^-e^-$ (LecNote18 p. 2, LecNote20 p. 3). LecNote19 p. 2's $\langle 0|Tb_{2'}b_{1'}b_1^\dagger b_2^\dagger|0\rangle$ has the fields $\Psi(x_2')\Psi(x_1')\bar\Psi(x_1)\bar\Psi(x_2)$. The scalar is internal, and the electron entering at x_1 leaves at x_1' or at x_2'. With $u_1=u_{s_1}(\mathbf{p}_1)$ and so on, $t=-(p_1-p_1')^2$, $u=-(p_1-p_2')^2$,

$$iT=\tfrac{1}{i}(ig)^2\Big[\frac{(\bar u_1'u_1)(\bar u_2'u_2)}{M^2-t}-\frac{(\bar u_2'u_1)(\bar u_1'u_2)}{M^2-u}\Big]. \tag{10.28.5}$$

The spinor orders $1'1\,2'2$ and $2'1\,1'2$ differ by one transposition, hence the minus: T is odd under $(\mathbf{p}_1',s_1')\leftrightarrow(\mathbf{p}_2',s_2')$.

$e^-e^+\to e^-e^+$ (LecNote18 p. 2). With $|i\rangle=b_1^\dagger d_2^\dagger|0\rangle$ and $|f\rangle=b_{1'}^\dagger d_{2'}^\dagger|0\rangle$, the fields of $\langle 0|Td_{2'}b_{1'}b_1^\dagger d_2^\dagger|0\rangle$ are $\bar\Psi(y_2')\Psi(x_1')\bar\Psi(y_1)\Psi(x_2)$. The direct term joins $y_1\to x_1'$ and $y_2'\to x_2$: scattering by φ exchange. The "exchange" term of LecNote18 p. 2 joins the incoming pair, $y_1\to x_2$, and the outgoing pair, $y_2'\to x_1'$: annihilation and pair creation, an s-channel graph. With $s=-(p_1+p_2)^2$, $t=-(p_1-p_1')^2$,

$$iT=\tfrac{1}{i}(ig)^2\Big[\frac{(\bar u_1'u_1)(\bar v_2v_2')}{M^2-t}-\frac{(\bar u_1'v_2')(\bar v_2u_1)}{M^2-s}\Big]. \tag{10.28.6}$$

The orders $1'1\,2\,2'$ and $1'2'\,2\,1$ differ by one transposition, so the relative minus stays though the particles differ. (Keeping every sign, the field order above adds an overall -1.)

$e^-e^+ \to \varphi\varphi$ (LecNote18 p. 1, LecNote20 p. 2). The arrow runs from the electron to the positron, and the φ's leave in either order: $iT = \frac{1}{i}(ig)^2\,\bar v_2[\tilde S(p_1-k_1)+\tilde S(p_1-k_2)]u_1$, the t- and u-channel graphs LecNote20 p. 2 draws. Its formula has $-\not{k}_{1,2}$ for $+\not{k}_{1,2}$, and m^2-s for $m^2-t=m^2+(p_1-k_1)^2$.